Chapter 11. Variable Selection

Consequences of omitting variables, search methods, selection criteria, Mallows' C_p

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With many candidate regressors we want a **parsimonious** model: enough variables to explain y well and predict accurately, but no more, since every extra variable adds variance to the estimates and predictions. This chapter quantifies the trade-off and describes search methods and criteria.

11.1 Consequences of omitting variables

Partition the **full model** as

$$\mathbf{y} = X\beta + \epsilon = X_k\beta_k + X_r\beta_r + \epsilon, \qquad \epsilon \sim N(\mathbf{0}, \sigma^2 I),$$

where X_k $(n \times (k+1))$ contains the intercept and k regressors and X_r $(n \times r,\ r = p-k)$ the others. The full-model estimator $\hat\beta = (\hat\beta_k', \hat\beta_r')'$ and $\hat\sigma^2 = SSE/(n-p-1)$ are compared with the **reduced model** estimators

$$\tilde\beta_k = (X_k'X_k)^{-1}X_k'\mathbf{y}, \qquad \tilde\sigma^2 = \frac{\mathbf{y}'\{I - X_k(X_k'X_k)^{-1}X_k'\}\mathbf{y}}{n-k-1}.$$

Theorem 11.1

Let $A = (X_k'X_k)^{-1}X_k'X_r$ (the **alias matrix**) and $H_k = X_k(X_k'X_k)^{-1}X_k'$. If the full model is true:

(a) $E\tilde\beta_k = \beta_k + A\beta_r$ and $\operatorname{Var}\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$; so $\operatorname{MSE}(\tilde\beta_k) = E(\tilde\beta_k - \beta_k)(\tilde\beta_k - \beta_k)' = \sigma^2(X_k'X_k)^{-1} + A\beta_r\beta_r'A'$.

(b) $E\tilde\sigma^2 = \sigma^2 + \dfrac{\beta_r'X_r'(I-H_k)X_r\beta_r}{n-k-1} \geq \sigma^2$.

(c) For predicting $E(y) = \mathbf{x}'\beta = \mathbf{x}_k'\beta_k + \mathbf{x}_r'\beta_r$ by $\tilde y = \mathbf{x}_k'\tilde\beta_k$:

$$\operatorname{MSE}(\tilde y) = \sigma^2\mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + (\mathbf{x}_k'A\beta_r - \mathbf{x}_r'\beta_r)^2.$$

Proof. (a) $E\tilde\beta_k = (X_k'X_k)^{-1}X_k'(X_k\beta_k + X_r\beta_r) = \beta_k + A\beta_r$; the variance is as usual; MSE $=$ Var $+$bias $\cdot$ bias$'$. (b) Theorem 5.4(a) with $I - H_k$, whose trace is $n-k-1$ and which annihilates $X_k\beta_k$. (c) MSE $=$ Var $+$bias^2 with bias $\mathbf{x}_k'(\beta_k + A\beta_r) - \mathbf{x}'\beta$. ■

So omitting relevant variables biases the remaining coefficients (unless $X_k'X_r = 0$) and inflates the residual variance estimate. In exchange the variance decreases:

Theorem 11.2

(a) $\operatorname{Var}(\hat\beta_k) - \operatorname{Var}(\tilde\beta_k)$ is nonnegative definite.

(b) $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{Var}(\mathbf{x}_k'\tilde\beta_k)$.

(c) If $\operatorname{Var}(\hat\beta_r) - \beta_r\beta_r'$ is nonnegative definite, then $\operatorname{Var}(\hat\beta_k) - \operatorname{MSE}(\tilde\beta_k)$ is nonnegative definite and $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k)$.

Proof. By the partitioned-inverse formula, with $B = X_r'(I-H_k)X_r$ (nonsingular),

$$\sigma^{-2}\operatorname{Var}(\hat\beta) = (X'X)^{-1} = \begin{pmatrix} (X_k'X_k)^{-1} + AB^{-1}A' & -AB^{-1} \\ -B^{-1}A' & B^{-1} \end{pmatrix}.$$

(a) $\operatorname{Var}\hat\beta_k - \operatorname{Var}\tilde\beta_k = \sigma^2 AB^{-1}A' \geq 0$. (b) With $\mathbf{x} = (\mathbf{x}_k', \mathbf{x}_r')'$ and $\mathbf{u} = A'\mathbf{x}_k - \mathbf{x}_r$,

$$\sigma^{-2}\operatorname{Var}(\mathbf{x}'\hat\beta) = \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + \mathbf{u}'B^{-1}\mathbf{u} \geq \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k.$$

(c) $\operatorname{Var}\hat\beta_r = \sigma^2B^{-1}$, so the assumption reads $\sigma^2B^{-1} \geq \beta_r\beta_r'$. Then $\operatorname{Var}\hat\beta_k - \operatorname{MSE}(\tilde\beta_k) = A(\sigma^2B^{-1} - \beta_r\beta_r')A' \geq 0$, and similarly $\operatorname{Var}(\mathbf{x}'\hat\beta) - \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k) = \mathbf{u}'(\sigma^2B^{-1} - \beta_r\beta_r')\mathbf{u} \geq 0$. ■

In words: dropping variables whose coefficients are small relative to their standard errors (so that $\beta_r \beta_r' \leq \operatorname{Var} \hat{\beta}_r$) reduces the mean squared error of estimation and prediction, even though it introduces bias. Conversely, if the reduced model is true ($\beta_r = \mathbf{0}$), both estimators are unbiased and the reduced one has smaller variance, so including irrelevant variables only costs precision.

11.2 Search methods

With p candidate regressors there are 2^p subsets (each variable in or out), so exhaustive search becomes infeasible for large p ($2^{50} \approx 10^{15}$).

All possible regressions (best subset). Fit all 2^p models. For each size k, keep the best model by R_k^2 (equivalently by SSE_k or MSE_k, since these are monotone in each other for fixed k), then compare the best models across k with a criterion that penalizes size (Section 11.3).

Backward elimination.

(1) Fit the full model.

(2) Compute the partial F statistic of each variable in the model.

(3) Let F_L be the smallest, and F_C the critical value at level α_{drop}.

(4) If $F_L < F_C$, delete that variable, refit and return to (2); otherwise stop.

This needs at most $p(p+1)/2$ fits.

Forward selection. Start with no regressors. At each stage add the variable that increases R^2 the most (equivalently, has the largest partial F given those already in), provided its partial F exceeds the critical value at level α_{add}; otherwise stop.

Stepwise selection. Forward selection, but after each addition the variables already in the model are re-examined by partial F tests and removed if they fall below the α_{drop} threshold (with $\alpha_{\text{drop}} \geq \alpha_{\text{add}}$ to avoid cycling). A variable that was useful early can become redundant once others enter.

The methods need not agree, and none guarantees the best subset. The levels used are not true significance levels, because of the repeated testing.

Example 11.1

In Hald's classical cement data ($n = 13$, four ingredients $x_1, \ldots, x_4$), x_2 and x_4 are highly correlated ($x_1+x_2+x_3+x_4 \approx 100$). Backward elimination ($\alpha = 0.05$) removes x_3 (partial $F = 0.02$) and then x_4 (partial $F = 1.86 < F_{0.05}(1, 9) = 5.12$), ending with $\{x_1, x_2\}$ ($R^2 = 0.979$). Forward selection enters x_4 first (largest single $R^2 = 0.675$), then x_1, and stops because x_2 has partial $F = 5.03 < 5.12$, ending with $\{x_1, x_4\}$ ($R^2 = 0.972$). Stepwise selection with $\alpha_{\text{add}} = 0.10$ enters x_2 and then removes x_4 (partial $F = 1.86$), ending with $\{x_1, x_2\}$. Because x_2 and x_4 carry nearly the same information, different routes end at different, almost equally good models.

11.3 Selection criteria

For a model with k regressors (plus intercept):

(1) residual mean square $MSE_k = SSE_k/(n-k-1)$;

(2) $R_k^2 = 1 - SSE_k/SST$;

(3) adjusted R^2: $R_{a,k}^2 = 1 - \dfrac{n-1}{n-k-1}(1-R_k^2) = 1 - \dfrac{MSE_k}{SST/(n-1)}$;

(4) Mallows' $C_k = \dfrac{SSE_k}{\hat\sigma^2} + 2(k+1) - n$, with $\hat\sigma^2$ the MSE of the full model.

R_k^2 never decreases when a variable is added, so it can only compare models of the same size. MSE_k can increase when a variable is added (the loss of one degree of freedom may outweigh the decrease in SSE), and $R_{a,k}^2$ is a decreasing function of MSE_k: maximizing R_a^2 is minimizing MSE_k.

Mallows' C_p. We judge the reduced model by the total standardized mean squared error of its fitted values at the n design points:

$$\Gamma_k = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{MSE}(\tilde y_i), \qquad \tilde y_i = \mathbf{x}_{ki}'\tilde\beta_k.$$

Theorem 11.3

$$\Gamma_k = (k+1) + \frac{\beta_r' X_r'(I-H_k)X_r\beta_r}{\sigma^2} = \frac{E(SSE_k)}{\sigma^2} + 2(k+1) - n.$$

Hence C_k estimates Γ_k, and if the reduced model is correct ($\beta_r = \mathbf{0}$), $E(SSE_k) = (n-k-1)\sigma^2$ and $C_k \approx k+1$.

Proof. By Theorem 11.1(c), the variance part sums to

$$\sum_i \mathbf{x}_{ki}'(X_k'X_k)^{-1}\mathbf{x}_{ki} = \mathrm{tr}\Big\{(X_k'X_k)^{-1}\sum_i \mathbf{x}_{ki}\mathbf{x}_{ki}'\Big\} = \mathrm{tr}\{(X_k'X_k)^{-1}X_k'X_k\} = k+1.$$

The bias vector over the design points is $X_kA\beta_r - X_r\beta_r = -(I-H_k)X_r\beta_r$, with squared length $\beta_r'X_r'(I-H_k)X_r\beta_r$. By Theorem 11.1(b), this equals $E(SSE_k) - (n-k-1)\sigma^2$. Substituting gives the second expression. ∎

So we look for models with small C_k and C_k close to $k+1$; a C_k well above $k+1$ signals bias from omitted variables. For the full model, $C_p = p+1$ identically.